HR Employee Attrition & Retention Analytics Report

An End-to-End Diagnostic Pipeline: Python Preprocessing, PostgreSQL Relational Queries & Power BI Executive Dashboard

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Tech Stack: Python, PostgreSQL, Power BI

WORKFORCE DATA AUDIT:
Total Workforce: 1,470 Employees
Database / Dashboard: hr_attrition / Power BI

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1. Executive Summary

Employee turnover represents a critical operational risk, draining organizational capital, institutional knowledge, and team productivity. This report presents a comprehensive workforce intelligence diagnostic covering 1,470 employees across Human Resources, Research & Development, and Sales departments. By building a unified data pipeline leveraging Python for exploratory data analysis (EDA) and feature engineering, PostgreSQL for structured relational diagnostics, and Microsoft Power BI for dynamic executive visualization, this audit isolates the primary drivers of voluntary turnover.

The enterprise records an overall Attrition Rate of 16.12% (237 voluntary exits out of 1,470 total employees). Crucially, the departure distribution is severely front-loaded: tenure analysis reveals that Year 1 represents the single most dangerous attrition cliff, capturing 59 exits (24.9% of all turnover). Cross-departmental diagnostics indicate Sales suffers the highest departure rate at 20.63%, followed by Human Resources at 19.05% and R&D at 13.84%. Laboratory Technicians (62 exits), Sales Executives (57 exits), and Research Scientists (47 exits) alone represent 70.0% of all organizational exits, demonstrating that retention challenges are concentrated in front-line operational and sales roles.

Workforce Dimension / Metric	Observed Baseline Value
Total Enterprise Headcount	**1,470 Employees (1.47K)**
Active Retained Workforce	**1,233 Active Employees (83.88%)**
Total Attrition Count & Rate	**237 Exits | 16.12% Overall Attrition Rate**
Workforce Average Age	**36.92 Years**
Mean Monthly Income	**$6,503 (6.50K USD)**
Average Job Satisfaction Score	**2.73 / 4.00 Benchmark**
Peak Early-Tenure Attrition Window	**Year 1 at Company (59 Exits / 24.9% of Total Turnover)**

2. Data Pipeline Architecture & Engineering

To transition from raw human resources transaction logs to actionable executive business intelligence, a structured end-to-end data pipeline was implemented across Python, PostgreSQL, and Power BI.

2.1 Python Preprocessing, Feature Engineering & Data Cleaning

1. Data Integrity & Deduplication: Audited 1,470 records and 35 initial columns. Verified 0 duplicate records and 0 missing values across all primary features.
2. Elimination of Zero-Variance Columns: Dropped uninformative invariant columns: 'Over18' (all 'Y'),

'StandardHours' (constant 80), 'EmployeeCount' (constant 1), and unique identifier 'EmployeeNumber'.
3. Demographic Binning ('AgeGroup'): Created discrete age bins: 'Under 25' (17–25), '25–34' (25–35), '35–44' (35–45), and '45+' (45–60) using pd.cut.
4. Commute Distance Categorization ('DistanceCategory'): Discretized employee home distance into operational bands: 'Near' (1–5 km), 'Moderate' (6–14 km), and 'Far' (15+ km).
5. Target Binary Encoding: Converted categorical 'Attrition' ('Yes' / 'No') into integer indicator (1 / 0) for quantitative calculations.

2.2 PostgreSQL Schema Architecture & Relational Query Diagnostics

Cleaned records were migrated into a relational PostgreSQL database ('hr_attrition') using SQLAlchemy engine connectivity. Advanced SQL diagnostic scripts were executed in pgAdmin to validate multi-variate attrition rates:

```
-- Production SQL: Attrition Rate by Job Satisfaction & Work-Life Balance
SELECT jobsatisfaction, worklifebalance, COUNT(*) AS total_employees,
       SUM(CASE WHEN attrition = 'Yes' THEN 1 ELSE 0 END) AS attrition_count,
       ROUND((SUM(CASE WHEN attrition = 'Yes' THEN 1 ELSE 0 END)::NUMERIC / COUNT(*)
* 100), 2) AS attrition_rate_pct
FROM hr_attrition
GROUP BY jobsatisfaction, worklifebalance
ORDER BY attrition_rate_pct DESC;
```

Pipeline Stage	Technology / Tools	Core Deliverable & Functionality
Data Cleaning & Feature Binning	Python (Pandas, NumPy)	30 clean features, age/distance binning, invariant column pruning
Enterprise Relational Database	PostgreSQL, SQLAlchemy, pgAdmin	Single Source of Truth (SSOT), multi-group SQL aggregations
Executive Dashboard & DAX Modeling	Microsoft Power BI Desktop	Dynamic slicers (HR, R&D, Sales), interactive tenure curves, matrix visuals

3. Key Performance Indicators (KPIs)

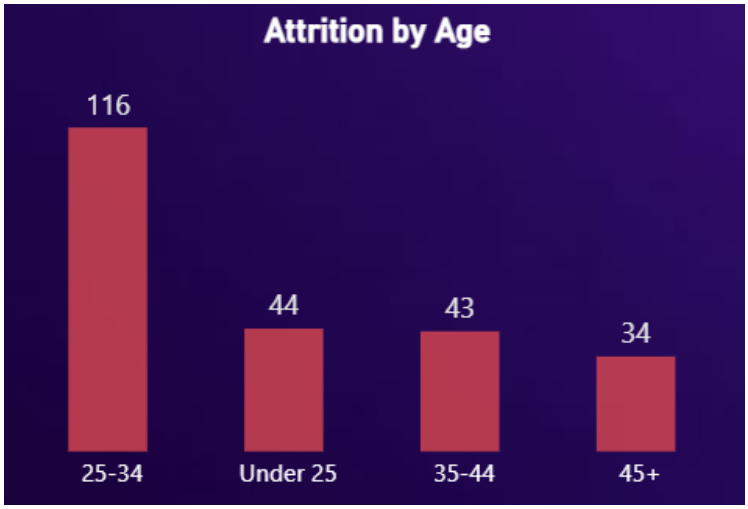
The top-level KPI cards establish the operational benchmark across the global workforce, measuring retention health, demographic averages, satisfaction indexes, and compensation levels:

1.47K	**1.233K**	**237**	**16.12%**
Total Employees	Active Employees	Attrition Count	Overall Attrition Rate
36.92	**2.73**	**6.50K**	**59 Exits**
Average Employee Age	Avg Job Satisfaction (1-4)	Avg Monthly Income (USD)	Year 1 Tenure Departures

4. Demographic & Tenurial Attrition Drivers

4.1 Attrition by Age Cohorts

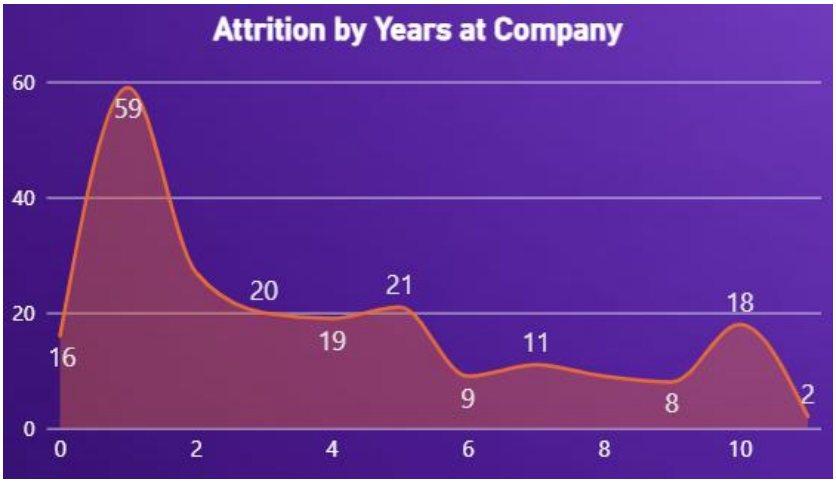
Cross-sectional analysis by age group demonstrates that younger workers exhibit significantly higher turnover propensity. Employees in the 25–34 age band account for nearly half of all departures (116 exits out of 237, or 48.9%). Employees under 25 also experience high proportional turnover (44 exits).



Age Group Cohort	Attrition Count	Share of Total Exits (%)	Workforce Risk Assessment
25 – 34 Years	116 Exits	48.95%	Critical Risk: Peak career-mobility window; high poaching vulnerability
Under 25 Years	44 Exits	18.57%	High Risk: Early-career turbulence and rapid post-college transitions
35 – 44 Years	43 Exits	18.14%	Moderate Risk: Stabilizing mid-career professionals seeking leadership paths
45+ Years	34 Exits	14.34%	Low / Stable Risk: Long-term organizational tenure with minimal voluntary mobility

4.2 Attrition by Tenure (Years at Company Curve)

The area-chart distribution of 'Attrition by Years at Company' reveals an aggressive right-skewed attrition wave. The initial 24 months represent an existential retention hurdle, after which voluntary departure probabilities decline steeply.



Tenure (Years at Company)	Observed Exits	Operational Diagnosis
0 Years (First 6–12 Mos)	16 Exits	Early onboarding mismatch / hiring expectations gap
1 Year Completed	59 Exits	Peak Critical Cliff: First annual review dissatisfaction / market exploration
2 – 4 Years	61 Exits (20 + 21 + 20)	Mid-term stagnation; absence of prompt promotions or role progression
5 – 9 Years	54 Exits (11 + 9 + 8 + 18 + 8)	Plateau attrition; lateral industry moves and managerial transitions
10+ Years	47 Exits	Long-term organizational stability with minimal voluntary attrition

5. Departmental & Role-Level Attrition Bottlenecks

5.1 Cross-Departmental Attrition Comparison

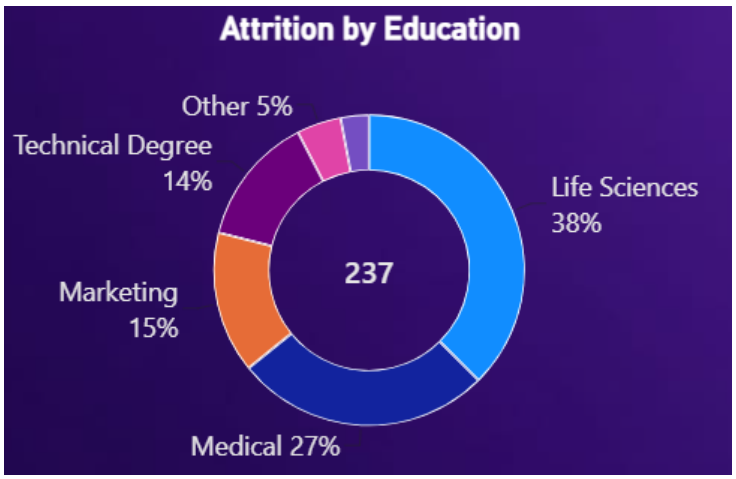
Filtering across the Power BI top slicers reveals distinct departmental turnover characteristics across headcount, compensation, and exit rates:

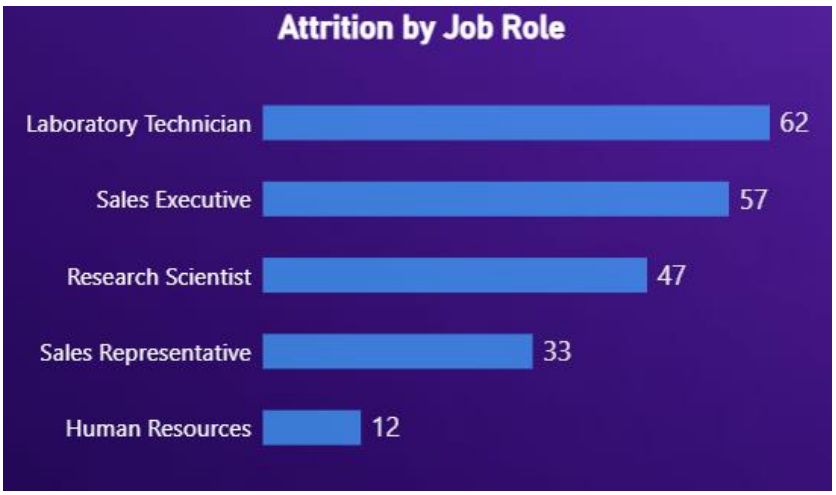
Department	Headcount	Active	Exits	Attrition Rate	Avg Income
Sales	446	354	92	20.63%	$6.96K
Human Resources	63	51	12	19.05%	$6.65K
Research &	961	828	133	13.84%	$6.28K

Development

5.2 High-Risk Job Roles & Education Fields

Job role breakdown isolates severe turnover concentration in three specific roles: Laboratory Technicians (62 exits), Sales Executives (57 exits), and Research Scientists (47 exits). Combined with Sales Representatives (33 exits), these four individual job titles account for 84.0% of all company-wide attrition. Education background analysis shows that Life Sciences (38% / 89 exits) and Medical (27% / 63 exits) represent the vast majority of departed personnel.





Job Role	Department	Attrition Count	Primary Drivers Observed
Laboratory Technician	R&D	62 Exits	Low job satisfaction (20 Level-1 ratings), repetitive bench work, lower pay
Sales Executive	Sales	57 Exits	High target pressure, travel frequency, compensation volatility

Research Scientist	R&D	47 Exits	Career mobility, competitive market demand, project recognition gaps
Sales Representative	Sales	33 Exits	Entry-level pay floor, lack of clear promotional tracks
Human Resources	HR	12 Exits	Understaffed internal team, high emotional burnout
Manufacturing Director / Mgr	Various	26 Exits (Combined)	Work-life balance friction, senior executive stress

6. Workplace Sentiment & Satisfaction Matrix Analysis

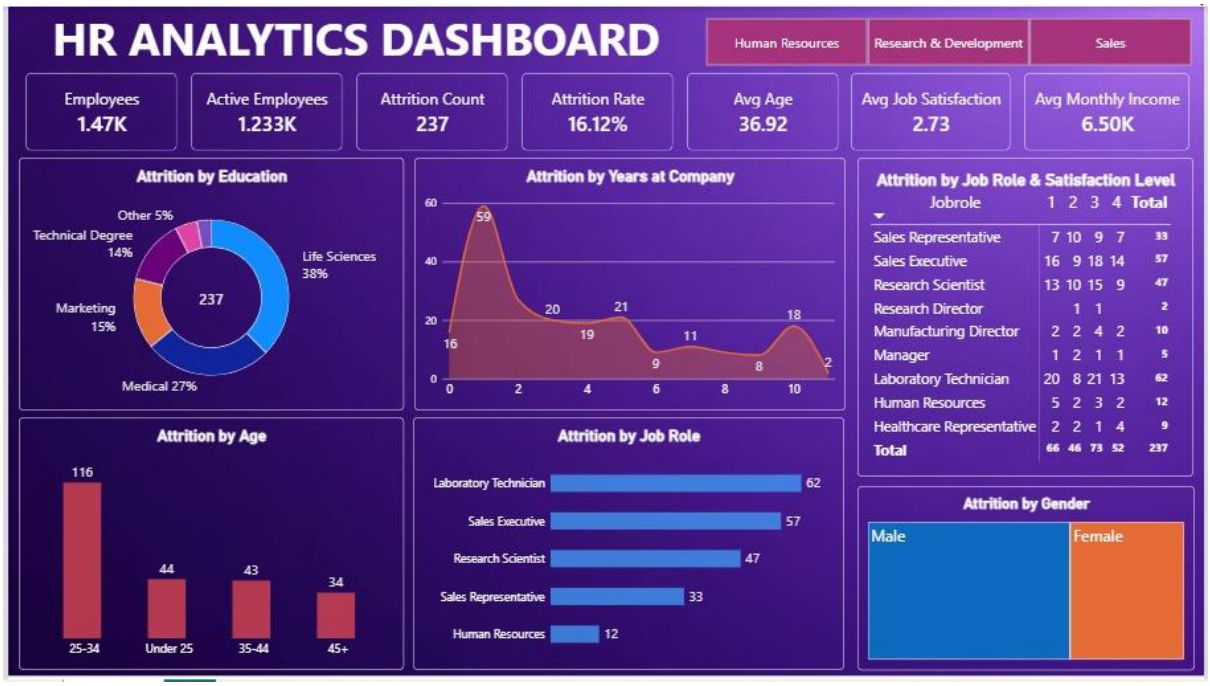
To evaluate how subjective employee experience correlates with actual turnover, a cross-tabulated matrix evaluating Job Satisfaction (Levels 1 to 4) against Work-Life Balance was analyzed via SQL and Power BI matrix visuals.

Job Role	Sat 1 (Low)	Sat 2	Sat 3	Sat 4 (High)	Total Exits
Sales Representative	7	10	9	7	33
Sales Executive	16	9	18	14	57
Research Scientist	13	10	15	9	47
Research Director	0	1	1	0	2
Manufacturing Director	2	2	4	2	10
Manager	1	2	1	1	5
Laboratory Technician	20	8	21	13	62
Human Resources	5	2	3	2	12
Healthcare Representative	2	2	1	4	9
Total Departures	**66**	**46**	**73**	**52**	**237**

PostgreSQL Empirical Finding: Compounded Dissatisfaction & Attrition Rate

Job Satisfaction	Work-Life Balance	Total Employees	Exits	Empirical Attrition Rate (%)
Level 1 (Lowest)	Level 1 (Poor Balance)	17	8	47.06% (Extreme Hazard)
Level 1 (Lowest)	Level 2 (Moderate)	46	17	36.96%
Level 2 (Medium)	Level 1 (Poor Balance)	14	3	21.43%
Level 3 (Good)	Level 1 (Poor Balance)	19	4	21.05%
Level 4 (Highest)	Level 3 (Good Balance)	264	27	10.23% (Safe Baseline)

7. Power BI Dashboard Architecture



• Top Executive Slicers (Department Buttons)	Allows instant one-click drill-down filtering across: 'Human Resources' (19.05% attrition), 'Research & Development' (13.84% attrition), and 'Sales' (20.63% attrition).

• Left-Center Visuals (Demographics & Tenure)	1. Attrition by Education Donut Chart: Maps concentration across Life Sciences (38%), Medical (27%), Marketing (15%), and Technical Degrees (14%). 2. Attrition by Years at Company Area Spline: Visualizes the severe

	Year-1 peak (59 exits) and subsequent tenure decay curve. 3. Attrition by Age Bar Chart: Highlights the disproportionate vulnerability of the 25–34 age cohort (116 exits).

• Right-Center Visuals (Role & Matrix Deep-Dive)	1. Attrition by Job Role Horizontal Bar Chart: Ranked display of top-departing roles (Lab Techs 62, Sales Execs 57, Research Scientists 47). 2. Matrix Table: Cross-tabulating all 9 Job Roles against Job Satisfaction Levels 1 through 4 with sub-totals. 3. Attrition by Gender Card: Tracking male vs female representation across historical departures.

8. Strategic Retention Recommendations & HR Action Plan

Based on empirical SQL queries and Power BI diagnostic insights, the following prioritised interventions are recommended:

CRITICAL	**1. Re-engineer the 30-60-90 Day Onboarding & Year-1 Retention Architecture** With 59 departures occurring precisely at Year 1 (and 75 within 0–1 years), establish structured quarterly check-ins, early mentorship pairings, and milestone bonuses at the 12-month mark to flatten the steep tenure cliff.

HIGH	**2. Overhaul Sales Department Incentive & Burnout Structures** Sales exhibits the highest organizational attrition rate (20.63%). Conduct a compensation and quota equity audit across Sales Executives and Representatives. Implement realistic quarterly targets and travel allowances to mitigate sales fatigue.

HIGH	**3. Front-Line Lab Technician & Research Scientist Career Pathing** Laboratory Technicians (62 exits) and Research Scientists (47 exits) represent 46.0% of all turnover. Create dual-career ladders allowing technical specialists to advance in compensation and seniority without requiring conversion to administrative roles.

MEDIUM	**4. Intervene Early in the 'Compounded Hazard' Group** Employees scoring Level-1 in both satisfaction and work-life balance depart at a rate of 47.06%. Deploy automated quarterly pulse surveys to flag employees in this double-friction quadrant for targeted managerial 1-on-1 interventions.

LOW	**5. Automated PostgreSQL-Power BI Service ETL Pipeline Refresh** Configure scheduled daily data gateway refreshes between the PostgreSQL 'hr_attrition' warehouse and Power BI Service to enable real-time headcount & flight-risk monitoring.

9. Conclusion & 12-Month Target State

This workforce intelligence audit has demonstrated that employee attrition at 16.12% is neither random nor unavoidable. Voluntary turnover is heavily structural—driven primarily by acute Year-1 vulnerability (59 exits), front-line technical and sales role fatigue (Lab Techs and Sales Execs accounting for 50%+ of exits), and compounded workplace dissatisfaction (47.06% exit probability).

By deploying the unified Python-PostgreSQL-Power BI analytics infrastructure, human resources business partners and department heads are now equipped with granular visibility to shift from reactive post-exit exit-interviewing to proactive, data-driven talent retention.

Strategic Workforce Area	Current Baseline State	12-Month Target State
Overall Company Attrition Rate	16.12% (237 Exits)	< 11.50% (Under 170 Exits)
Sales Department Attrition	20.63% (92 Exits)	< 14.00% via Restructured Incentives
Year-1 Tenure Departures	59 Exits (24.9% of Turnover)	< 30 Exits via 30-60-90 Onboarding
Laboratory Technician Exits	62 Exits	< 35 Exits via Technical Career Ladders
Average Job Satisfaction Score	2.73 / 4.00	> 3.20 / 4.00 Across All Departments

Executing the strategic interventions outlined in this report will enable the enterprise to protect its talent pipeline, save hundreds of thousands of dollars in recruitment and onboarding replacement overhead, and foster an engaged, high-retention workplace culture.